

Nathan Haley

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Mission Statement

I would love to pursue a role at a company where I can work on multi-disciplinary engineering challenges and meet amazing people. I am passionate about design, teamwork, and learning new things. I am a customer focused individual with keen attention to detail who enjoys being creative, troubleshooting, and making the world a better place.

Work & Project Experience

Holmberg Mechanical | Bellevue, WA

Sep 2020 -Sep 2021, June 2025-August 2025 (contract)

Project Engineer/Mechanical Engineer (Plumbing)

- **Reviewed** DIV 22 and 23 submittals and specs for equipment and commodities
- **Selected** main components for high pressure hydronic system (+300psi)
- **Issued** and managed RFIs for mechanical and plumbing scopes
- **Collaborated** with seismic, thermal exp., controls, and detailing subcontractors
- **Assisted** site superintendents with day-to-day and onsite QA/QC inspection tasks
- **Assisted** engineering and detailing with assorted tasks
- Contract work scope
 - **Designed** domestic plumbing water and waste/vent systems for high-rise building

PSR Mechanical | Seattle, WA

Sep 2021-Aug 2024, April 2025-May 2025 (contract)

Project Executive/Project Engineer

- **Performed** sales, estimating, and early stage management of various mechanical/HVAC TI and equipment replacement projects
- **Provided** sales, estimating, and full life cycle management and engineering of local and WA state mandated energy improvement ordinance projects
- **Developed** knowledge base regarding current and upcoming local and WA state mandated energy improvement ordinances and construction codes
- **Creation** of mechanical permit sets and permit applications for various TI projects
- **Calculation** of thermal and ventilation loads
- Contract work scope
 - See previous two bullets

Personal Goal Pursuit/Self Study

Aug 2024-March 2025

Software Engineering

- Full time self-study and **personal project development in the field of software engineering**. I have transitioned to part-time project development since March 2025
- Gained proficiency in **C#, Python, HTML, and CSS** programming languages. Some JavaScript as well
- Gained proficiency with **AI agentic tools** such as Github Copilot Agent Mode
- Gained proficiency in **Visual Studio Community/Code and Unity** programming environments
- Released **music player application** and **duct static pressure calculator** personal programming projects on customized website I built from scratch (see more details at nathanwhaley.com)
- Worked on data processing tasks for an **AI/machine learning project** involving the WA State Clean Building Performance Standard (CBPS). Project is unreleased currently
- Gained knowledge of common software engineering **design patterns** such as Model-View-Controller (MVC), Observer, Composite, State, Command, Singleton, and Proxy patterns that I have implemented across my released projects

O'Reilly Auto Parts | Bellevue, WA**March 2025-Present***Delivery Driver/Counter Sales*

- **Deliver** auto parts locally to commercial customers (i.e, mechanic shops and franchise locations) and return pickups in company vehicle
- **Counter sales**, basic automotive troubleshooting, and basic assistance (i.e, code scanning, battery replacements, headlight replacements, etc.)
- **Inventory stocking**

Rushing Company | Seattle, WA**April 2019-Mar 2020***Mechanical Project Engineer*

- **Designed** and modeled ventilation, thermal comfort, pressurization, and exhaust systems in Revit and system schematics in AutoCAD
- **Selected** fans, pumps, VRF FCUs, fire smoke dampers, louvers, and GRDs using vendor selection software and product catalogs
- **Calculated** thermal loads for dwelling, amenity, and kitchen spaces for high/mid- rise hotel/luxury apartment projects
- **Designed** Detailed Event Matrix for stairwell/hoistway pressurization operation during a fire/smoke event
- **Learned** about refrigeration piping considerations for VRF heat pump systems
- **Learned** about rational analysis for smoke control systems from NFPA 92
- **Learned** about 2015 Seattle Mechanical, Building, and Energy Codes

McKinstry Company | Seattle, WA**Oct 2018-Mar 2019***Electrical Detailer*

- **Detailed** electrical power and low voltage as-built sets in Bluebeam and Revit
- **Modeled** electrical construction and fabrication spools in Revit and AutoCAD
- **Automated** Revit tasks using Dynamo API and Python script nodes in Dynamo

Exotic Metals Forming Company | Kent, WA**Jul 2016-Jul 2018***Facilities Engineering Technician*

- **Drafted** shop floor layouts for production cells in AutoCAD
- **Drafted** install plans for critical assets (presses, machining centers, etc.)
- **Drafted** facility MEP system plans for new installation and permitting purposes
- **Designed** relational database for manufacturing/facilities asset management applications using browser based ERDPlus system modeling tool
- **Troubleshooted** HVAC issues with an HVAC SCADA/BMS system
- **Managed** HVAC maintenance service contracts for 5 facilities (100+ HVAC assets)
- **Managed** parking lot repaving project and published work specifications

Education & Training

AEE Certified Energy Manager (CEM)**Mar 2022 - Present****ICC M3 2018 Mechanical Plans Examiner****Apr 2021 - Present****LEED® Green Associate™****Nov 2019 - Present****Udemy.com Complete Python Bootcamp w/ Jose Portilla****Mar 2019 - Apr 2019**

- 12+ hrs of online instruction and 40+ hrs of hands on coding
- Notable projects include Tic-Tac-Toe, Blackjack, and a basic music player

Washington State Engineer in Training**Dec 2018-Present****Autodesk Certified Professional in Revit MEP: Mechanical****Mar 2017-Present****B.S., Mechanical Engineering w/ Materials Science Minor****Aug 2012- May 2016**

- Washington State University, 3.80 GPA, Magna Cum Laude
- Mentor in Mechanical and Materials Engineering Mentorship Program

Programming Skills

C#, Unity, Python, Github Copilot Agent Mode, OpenAI Codex, HTML, CSS, JavaScript, SQLite, EmailJS, Render Web Services, MATLAB, Visual Studio, Github, Jira, Excel VBA, Dynamo for Revit, Web Design, Application Design